side. Your well-trained crew and robots rush to their battle stations, you search vainly for the attacking warship—and then, responding to an appeal for her help, DAWN announces soothingly over the ship's speaker system, "Tikhii, tikhii; we are not being attacked. That was just a freak primordial black hole, evaporating and then exploding."[37]

"A what?!" you cry out.

"A primordial black hole, evaporating and then destroying itself in an explosion," DAWN repeats.

"Explain!" you demand. "What do you mean by *primordial?* What do you mean by *evaporating and exploding?* You're not making sense. Things can fall into a black hole, but nothing can ever come out; nothing can 'evaporate.' And a black hole lives forever; it always grows, never shrinks. There is no way a black hole can 'explode' and destroy itself. That's absurd."

Patiently as always, DAWN educates you. "Large objects—such as humans, stars, and black holes formed by the implosion of a star—are governed by the *classical* laws of physics," she explains, "by Newton's laws of motion, Einstein's relativity laws, and so forth. By contrast, tiny objects—for example, atoms, molecules, and black holes smaller than an atom—are governed by a very different set of laws, the *quantum* laws of physics.[38] While the classical laws forbid a normal-sized black hole ever to evaporate, shrink, explode, or destroy itself, not so the quantum laws. They demand that any atom-sized black hole gradually evaporate and shrink until it reaches a critically small circumference,about the same as an atomic nucleus. The hole, which despite its tiny size weighs about a billion tons, must then destroy itself in an enormous explosion. The explosion converts all of the hole's billion-ton mass into outpouring energy; it is a trillion times more energetic than the most powerful nuclear explosions that humans ever detonated on Earth in the twentieth century. Just such an explosion has now damaged our ship," DAWN explains.

"But you needn't worry that more explosions will follow," DAWN continues. "Such explosions are exceedingly rare because tiny black holes are exceedingly rare. The only place that tiny holes were ever created was in our Universe's big bang birth, fifteen billion years ago; that is why they are called *primordial* holes. The big bang created only a few such primordial holes, and those few have been slowly evaporating and shrinking